

RKHS Extrinsic Geometry and Interpolation — Summary

1. Core Idea:

RKHS studies subspaces $V \subset \mathbb{R}^n$ using their embedding (extrinsic geometry), not just abstract structure.

2. Intrinsic vs Extrinsic Geometry:

- Intrinsic: inner product structure.
- Extrinsic: how the space sits inside $\mathbb{R}^n$.

Two spaces can be intrinsically identical but extrinsically different.

3. Interpolation Problem:

Find $x \in V$ such that:

$$e_j^T x = 1$$

$\|x\|$ is minimum

4. Geometric Insight:

The optimal x must be orthogonal to all vectors $v \in V$ such that $e_j^T v = 0$.

5. Problem with Basis:

Using a basis is:

- Not unique
- Not continuous
- Not stable as V changes

6. RKHS Solution:

Use kernel vectors $k_1, \dots, k_n$ instead of basis.

7. Key Representation:

Any $x \in V$ can be written as:

$$x = K\alpha$$

8. Final Solution:

$$x = k_j / K_{jj}$$

9. Interpretation:

- k_i is the gradient of coordinate function.
- Solution aligns with direction of k_i .

10. Key Advantage:

Kernel representation varies continuously with V , leading to stable and elegant solutions.

Conclusion:

RKHS replaces arbitrary basis choices with canonical kernel-based representations, enabling smooth, geometrically meaningful solutions to interpolation problems.